



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Circular Linked List

CIRCULAR LINKED LIST

"In a round-robin CPU scheduling system, processes are executed cyclically. Such applications can be represented using a circular linked list. Write a C program to create a circular linked list and display all nodes by traversing the list exactly once. Write a C program to create a circular linked list and traverse all nodes exactly once.

Test Case 1

Input

10 → 20 → 30 → 40

Output

10 → 20 → 30 → 40 → Back to Head"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 struct Node* createNode(int value) {
10     struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
11     if (!newNode) {
12         printf("Memory allocation failed!\n");
13         exit(1);
14     }
15     newNode->data = value;
16     newNode->next = NULL;
17     return newNode;
18 }
19
20 void insertAtEnd(struct Node **head, int value) {
21     struct Node* newNode = createNode(value);
```

OUTPUT

```
20 -> 30 -> 40 -> 40 -> 40 -> 40 -> 40 -> 40 -> 40 -> 40 ->
Back to Head
```

INPUT

10 20 30 40

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